Smoothing in Financial Time Series

Core idea

Smoothing estimates a **latent state** (trend/level/volatility) by suppressing high-frequency noise

- **Latent trend / level -** trend level in *log-price*
- **Volatility / risk state** - volatility regime, variance level
- **Regime labels / turning points -** Bull vs bear," "trend break," "cycle phase" (state classification)
- **Signal construction -** slope of trend, residual volatility

- Smoothing mainly applies to state-estimation (trend/cycle/vol regime) but is not additive to return prediction.
  - Example : 30 returns are close to martingale + noise, smoothing will not add much value

- Critical constraint
  - **One-sided smoothing is: SAFE:** uses data available up to point t (AKA Causal)
  - **Two-sided smoothing is: LEAKY**: uses future data t+1, t+2,…. (AKA Centered)

Smoothing tools

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Objectives Map – One-sided (no-leakage) Smoothing

Objective	Input to smooth	Why it's the right input	Common one-sided methods (no leakage)
Trend / latent state (level, slow regime proxy)	**log(price)**	Stable scale; trend + residual is additive/interpretable;	• **EMA/SMA**, • **Kalman Local-LT**, • **LOWESS** • **SG** • **SSA**
Volatility / risk (variance / regime)	**r^2**, $\|r\|$, **realized variance (RV)**, or **log(RV)**	• Volatility clusters and is persistent • Smoothing improves stability of variance estimates	• **EWMA** on r^2, • SMA/EMA on RV • **Kalman** on log-variance

Smoothing Engines

Category	Method	Application
Moving average (MA)	Simple moving average **(SMA)**	Trailing mean over a fixed window; strong lag.
Moving average (MA)	Exponential moving average **(EMA/EWMA)**	Trailing mean with exponentially decaying weights; less lag than SMA.
Moving average (MA)	**Rolling median / trimmed mean**	Robust smoother that downweights spikes/outliers.
State space	**Kalman Local Linear Trend** (LLT)	Trend = level + slope state; adapts to changing regimes with uncertainty.
State space	**Local level / DLM**	Trend = latent level; simplest Kalman trend model.
State space	**Kalman on log-variance**	Models latent volatility state (often on $\log(\sigma^2)$), smooths vol regimes.
Regression / local fit	**LOWESS/LOESS**	Local weighted regression; captures nonlinear slow drift.
Regression / local fit	**Savitzky–Golay (SG)**	Local polynomial fit; preserves slopes/curvature better than MA.
Parametric / ETS	**Holt / Holt-Winters**	Exponential smoothing with trend (and seasonality if used).
Penalized trend	**HP filter**	Trend–cycle decomposition via smoothness penalty.
Subspace / spectral	**SSA (SVD)**	Low-rank Hankel/SVD reconstruction; separates slow components.
ML representation	**Autoencoder (AE) denoiser**	Learns nonlinear compression; reconstruction can remove high-freq noise.

Smooth What

Objective	Example variables	Common one-sided smoothers
Trend level / latent state (30d+ for markets; months+ for macro)	Equity index level (SPY/SPGI), sector ETFs	**EMA/SMA on logp, Kalman LLT filter, trailing LOWESS/SG, rolling SSA**
Trend slope / momentum feature	Equity/credit indices, FX spot	**EMA/Kalman trend + slope, SG (trailing) derivative, rolling SSA**
Regime / bull–bear (medium horizon)	Index level, breadth proxies	**EMA/Kalman trend, residual z-score, drawdown features**
Short-horizon return prediction (next day–few weeks)	Daily returns on SPY/SPGI	**Trend-based features from log(price) (slope, residual), not smoothed returns**
Volatility / risk / stress	Equity vol, credit spread vol	**EWMA on r^2, SMA/EMA on RV, Kalman on $\log(\sigma^2)$**
Inflation trend / inflation regime	CPI, Core CPI, PCE	**EMA/SMA on YoY, Kalman LLT on YoY, trailing LOWESS**
Inflation "momentum" / turning points	CPI YoY / MoM	**Short EMA on YoY-diff; Kalman LLT with small process noise**
Output/growth trend	Real GDP, Industrial Production (IP), Retail sales	**EMA/SMA on growth rate, Kalman/DLM, trailing LOWESS**
Business cycle / gap proxy	GDP, IP, unemployment rate	**Rolling one-sided HP-like (refit per window), rolling SSA**
Policy rate / rate trend	Fed funds, Bank Rate, ECB deposit	**EMA/SMA on level, Kalman local level/LLT**
Yield curve / term structure trend	2y/10y yields, curve slope (10y–2y)	**EMA on spread, Kalman on level/slope states**
Labour market trend	Unemployment rate, payrolls	**EMA/SMA, Kalman local level**

Smoothing Engine Application

General Note

- Many smoothing “engines” can be run in either **one-sided** (real-time) or **two-sided** (cantered) mode
- Separate: descriptive smoothing for insight vs. causal smoothing for forecasting/backtesting

Core distinction

- **Descriptive smoothing (insight/communication):** ex-post, can use cantered filters.
- **Forecasting/backtesting (decision-grade):** must be one-sided to avoid leakage.

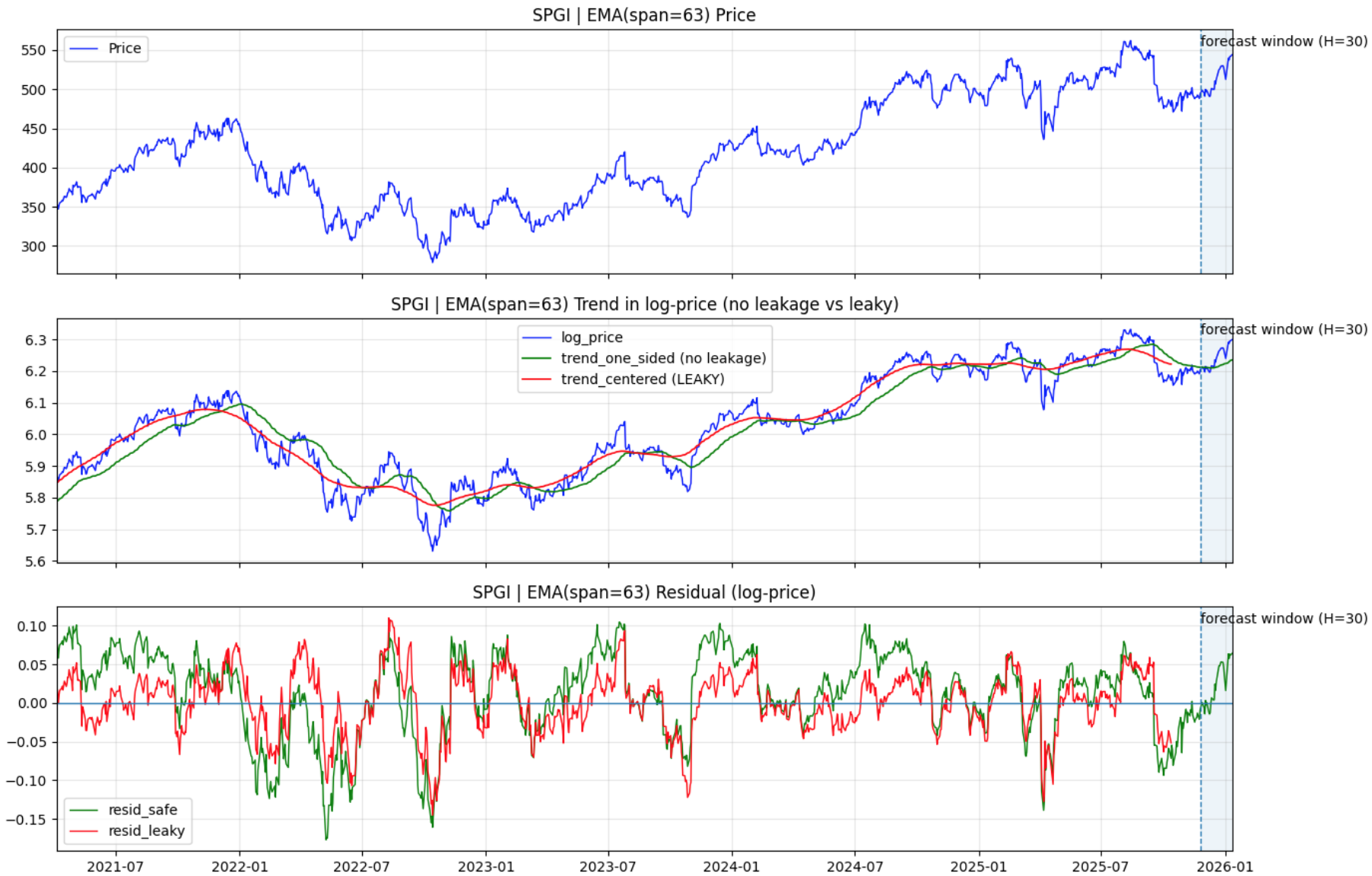
One-Sided (Real-Time) Smoothing — “Decision-Grade”

- Building features for **forecasting**, **signals**, **backtesting**, or any **walk-forward** evaluation.
- Prevents **forward leakage** and overly optimistic performance.
- You need a value that could have been is “available-at-t” - trend/residual/slope features.

Cantered Smoothing application

- **Post-model diagnostics and communication:** cleaner overlays and narrative plots.
- **Offline decomposition:** separating low-frequency trend/cycle from noise.
- **Ex-post reference labels:** turning points, historical regimes, benchmark “trend truth.”

Smoothing Engine Application - Example



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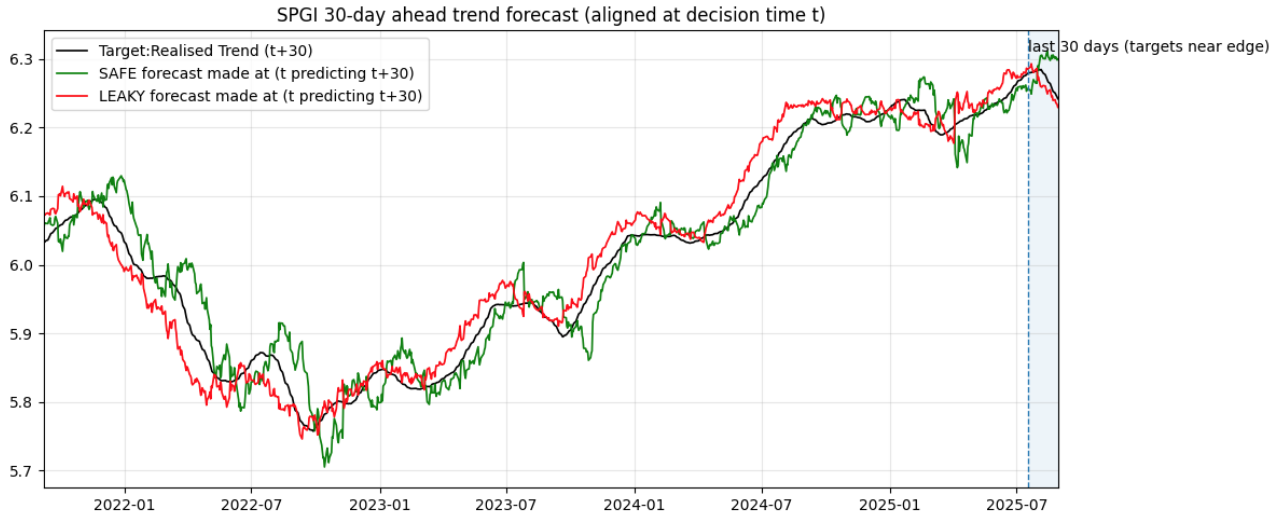
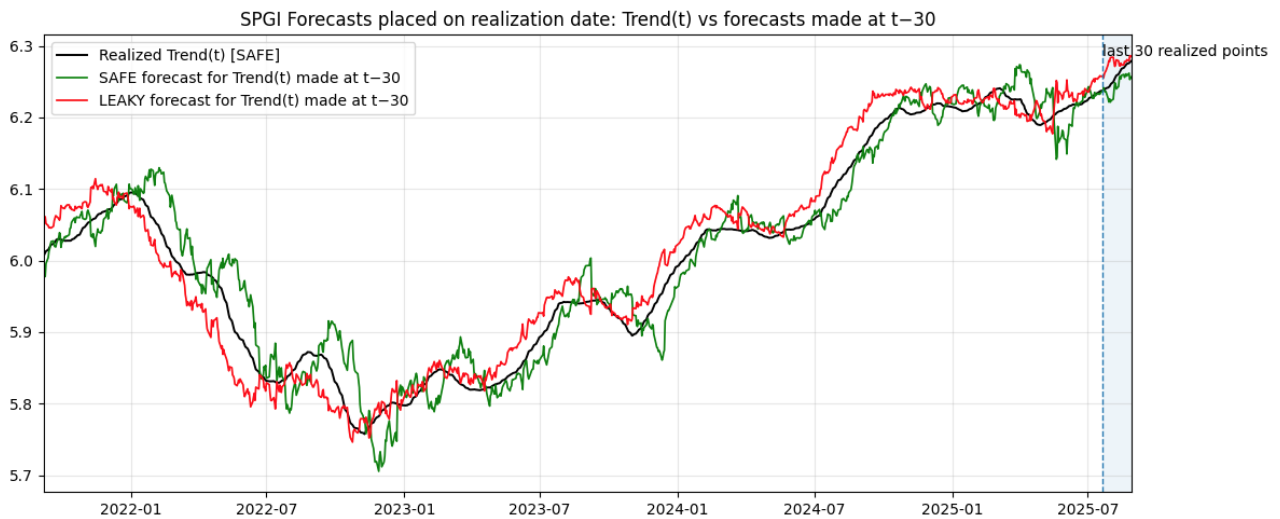
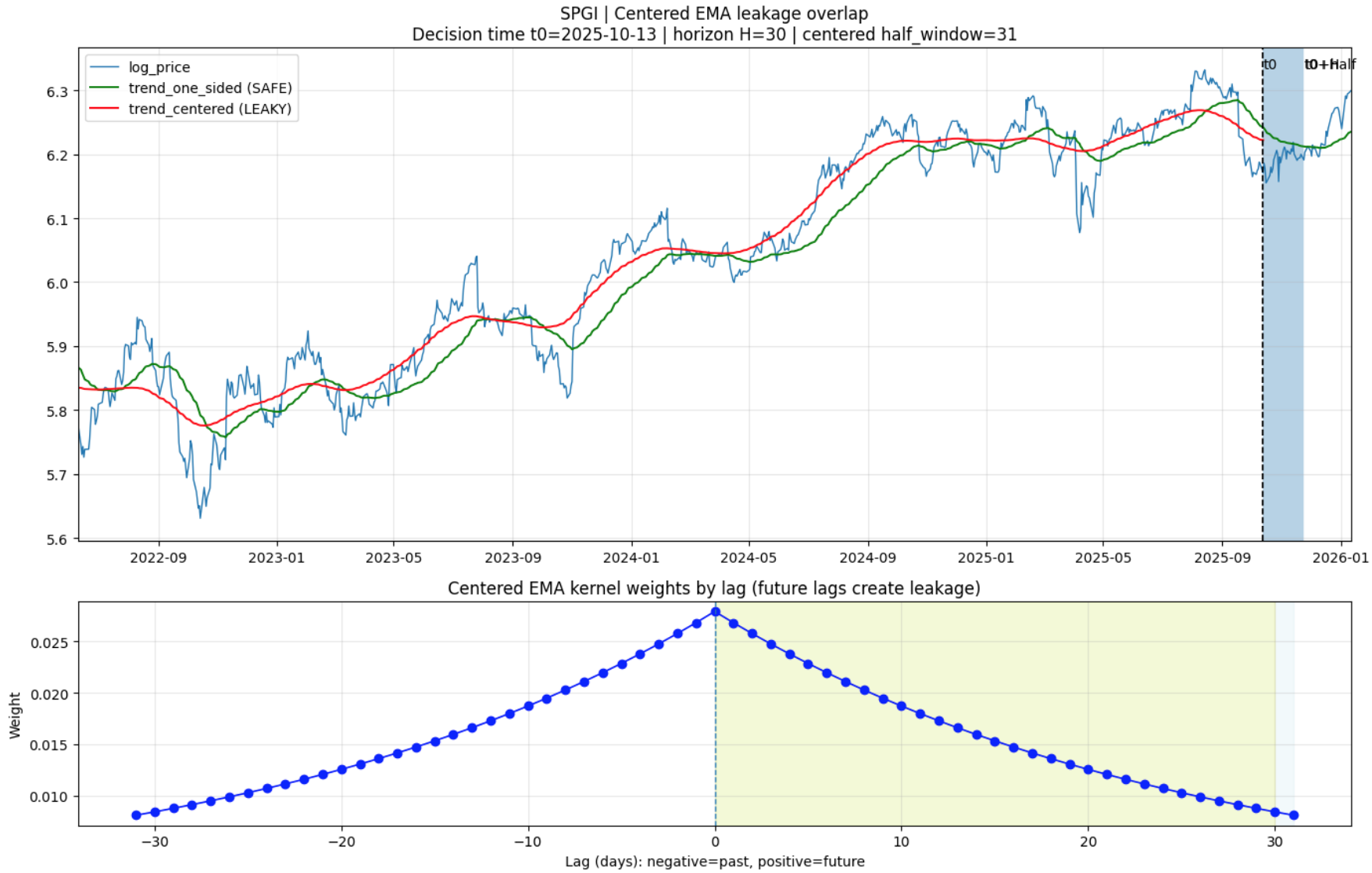




========== 30-day ahead trend forecast metrics (target = SAFE trend at t+30) =========

SAFE features : {'**MSE**': 0.0016414733107874154, '**MAE**': 0.028954241281127755, 'Corr': 0.9993683270943055, 'n_test': 5992}
LEAKY features: {'**MSE**': 0.0004668769267320019, '**MAE**': 0.01623216677852311, 'Corr': 0.9998154580427061, 'n_test': 5868}

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